

Physics A (H156, H556)

Exploring-Magnetism-Q+MS KSA Physics

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:

Additional resources may be supplied with this paper.

Other materials required:

- Pencil
- Ruler (cm/mm)

Duration: Not set

Candidate forename		Candidate surname	
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Centre number						Candidate number				
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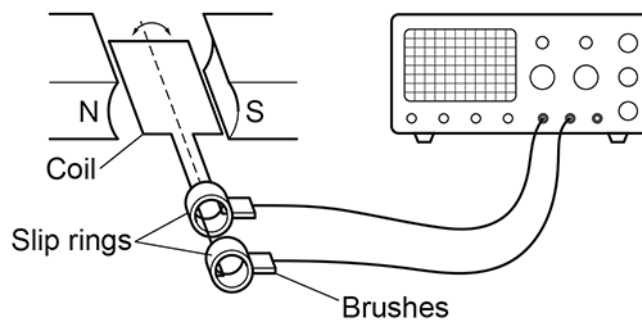
INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets [] at the end of each question or part question.
- The total number of marks for this paper is **79**.
- The total number of marks may take into account some 'either/or' question choices.

1(a) The diagram shows a simple a.c. generator connected to an oscilloscope (CRO).



Students turn the coil at a steady rate and an a.c. signal is displayed on the oscilloscope screen, as shown in Fig. 21.1.

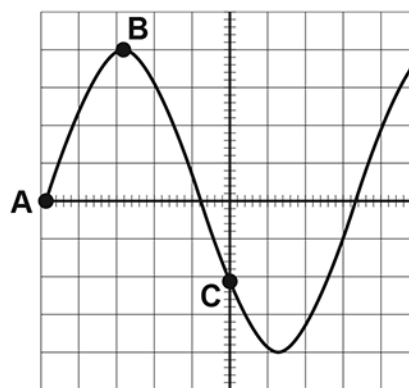
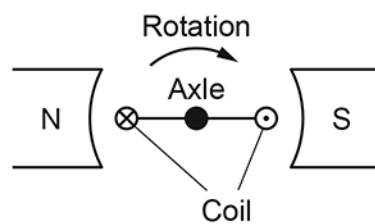


Fig. 21.1

Which position on the signal, A, B or C, corresponds to the generator position shown below?



view from end of coil

Tick **one** box.

A	☐
B	☐
C	☐

[1]

(b) Draw on Fig. 21.1 to show how the signal would change if the coil is rotated at a steady, faster rate.

[2]

- (c) The coil is rotated 45° clockwise from the position shown in (a) in a time of 0.13 s.
The magnetic field between the poles of the magnet is uniform and is 25 mT.
The coil has 120 turns.
The coil is 8.0×10^{-2} m long and 4.0×10^{-2} m wide.

- i. Calculate the change in flux linkage.

Change in flux linkage = Wb [2]

- ii. Calculate the average induced electromotive force (e.m.f.) $\mathcal{E}$ across the contacts of the coil.

$\mathcal{E}$ = V [2]

(b) A simple laminated iron-core transformer takes mains voltage 230 V, 50 Hz into the primary coil. The output voltage from the secondary coil is 5.0 V. The primary coil has 920 turns.

i. State *Faraday's law*.

-----[1]

ii. Show that the number of turns on the secondary coil is 20.

[2]

iii. At one particular instant, the output voltage from the transformer is 3.4 V.

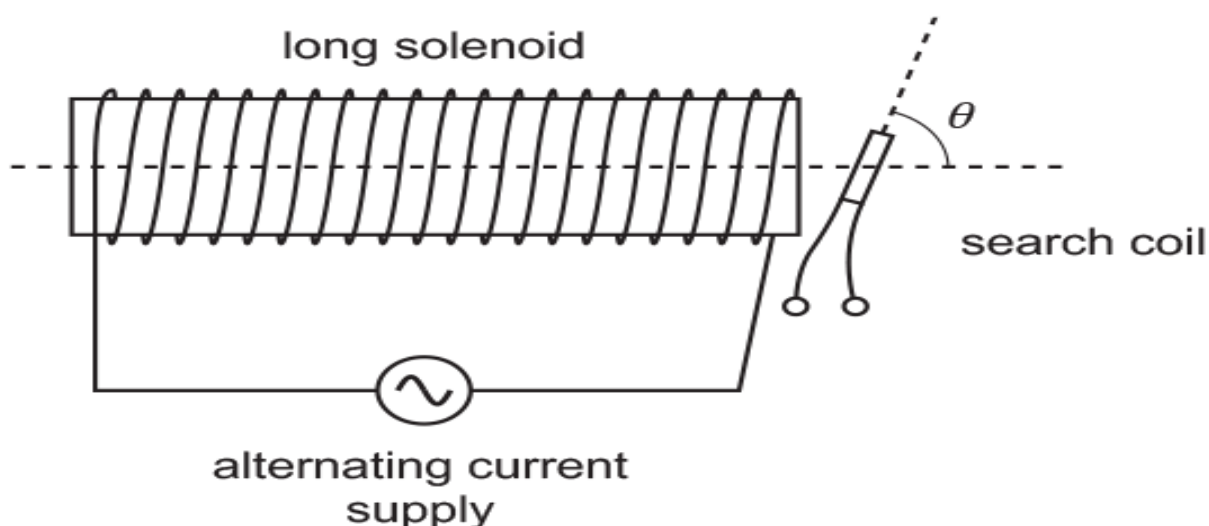
Calculate the change in magnetic flux experienced by the secondary coil in a short time interval of 1.2 ms and state its unit.

Assume that the output voltage from the transformer remains constant at 3.4 V over this time interval.

$\Delta\phi =$ unit [4]

3(a)

A student is carrying out an experiment using a search coil.



A long solenoid is connected to an alternating current supply.

The search coil is placed at one end of the solenoid. The plane of the search coil is tilted such that it makes an angle θ with the central axis of the solenoid. The maximum alternating induced electromotive force (e.m.f.) across the ends of the search coil is E_0 .

- i. Name an instrument that can be used to determine E_0

[1]

- ii. The equation for E_0 is:

$$E_0 = KI_0 ANf \sin \theta$$

where I_0 = maximum current in the solenoid, A = cross-sectional area of the search coil, N = number of turns of the search coil, f = frequency of the alternating current in the solenoid and $K = 4.0 \times 10^{-3} \text{ VA}^{-1} \text{ m}^{-2} \text{ s}$.

The magnitude of the induced e.m.f. in the search coil can be determined using Faraday's law of electromagnetic induction:

e.m.f. = rate of change of magnetic flux linkage

In the experiment, angle θ is changed and E_0 measured.

Suggest the quantity, or quantities, in the equation $E_0 = KI_0 ANf \sin \theta$ linked to

- 1 the 'rate' part of the law

[1]

- 2 the 'change of magnetic flux linkage' part of the law.

iii. The student plots a straight-line graph of E_0 against $\sin \theta$.

Determine f , including the absolute uncertainty. Write your value of f to 2 significant figures.

$$I_0 = (8.0 \pm 0.2) \text{ A}$$

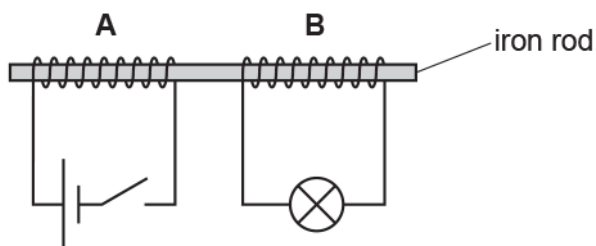
$$A = (7.8 \pm 0.1) \times 10^{-5} \text{ m}^2$$

$$N = 5000$$

$$\text{gradient of line} = KI_0 ANf = (0.62 \pm 0.03) \text{ V}$$

$$f = \dots\dots\dots \pm \dots\dots\dots \text{ Hz [4]}$$

(b) The diagram below shows two insulated-copper coils **A** and **B** connected in circuits.



Both coils are individually wrapped around the same iron rod.

Coil **A** is connected to a cell and a switch. Coil **B** is connected to a filament lamp.

The switch is initially closed and the lamp is off.

The switch is then opened. The lamp flashes on for a brief time, and then remains off.

Explain these observations in terms of magnetic flux.

.....

.....

.....

.....

.....

[3]

4

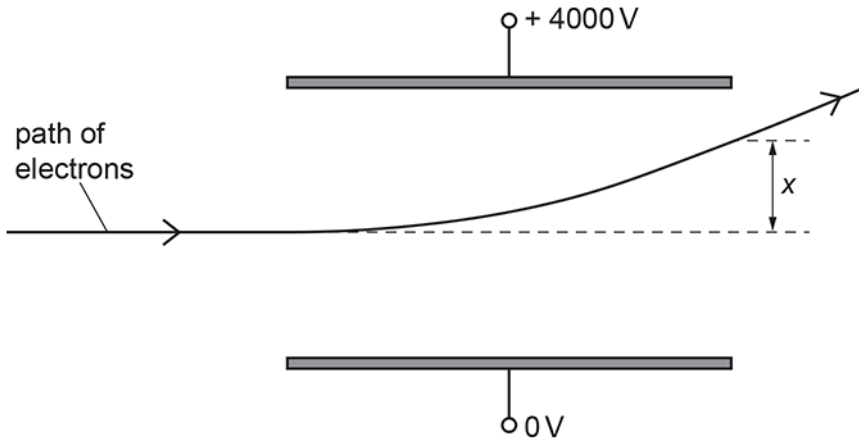


Fig. 24

Beta-minus particles (electrons) emitted from a radioactive source have a range of speeds.

Describe and explain how a uniform magnetic field can be applied in the space between the charged plates to select beta-minus particles with a specific speed. No calculations are required.

Blank lined paper for writing.

5(a) The diagram below shows the top-view of a long current-carrying wire.

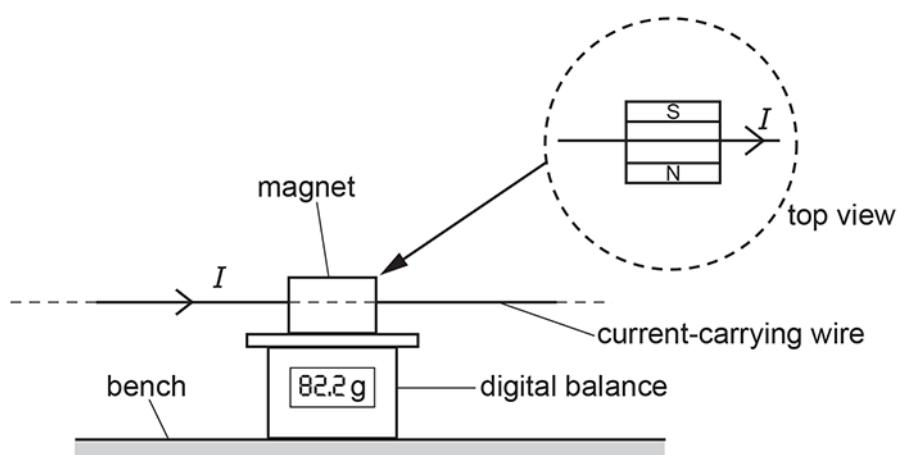


The direction of the current in the wire is into the plane of the paper.

Draw at least **three** field lines to indicate the magnetic field pattern around this wire.

[2]

- (b) The arrangement shown in the diagram below is used to determine the magnetic flux density between the poles of a permanent magnet



The magnet is placed on the digital balance. The current-carrying wire is horizontal and at right angles to the magnetic field between the poles of the magnet. The wire is fixed.

The following results are collected.

- length of the wire in the uniform field of the magnet = 6.0 ± 0.2 cm
- balance reading with no current in wire = 80.0 g
- balance reading with current in wire = 82.2 g
- current in wire = 5.0 ± 0.1 A

Calculate the magnetic flux density B , including the absolute uncertainty.

Ignore the absolute uncertainty in the balance readings.

Write your value for B to 2 significant figures and the absolute uncertainty to 1 significant figure.

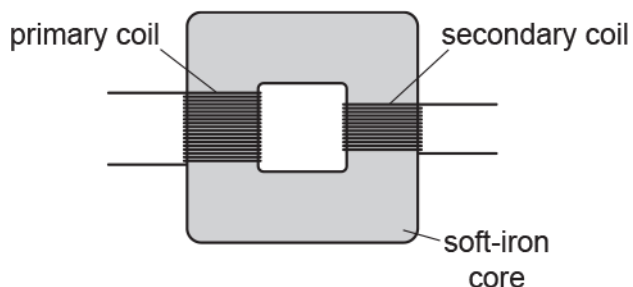
$B = \dots\dots\dots \pm \dots\dots\dots \text{ T [4]}$

6(a)

- i. State Faraday's law of electromagnetic induction.

[1]

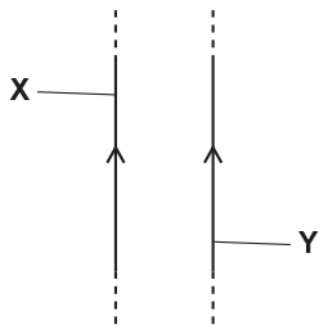
- ii. The diagram below shows a simple transformer constructed by a student.



Describe how the student can do an experiment in the laboratory to show that the maximum electromotive force (e.m.f.) E induced in the secondary coil is directly proportional to the number of turns N on the secondary coil.

[3]

(b) The diagram below shows two long vertical current-carrying wires *X* and *Y*.



The direction of the current in each wire is the same.

Explain why wire *Y* experiences a force and deduce the direction of this force.

[3]

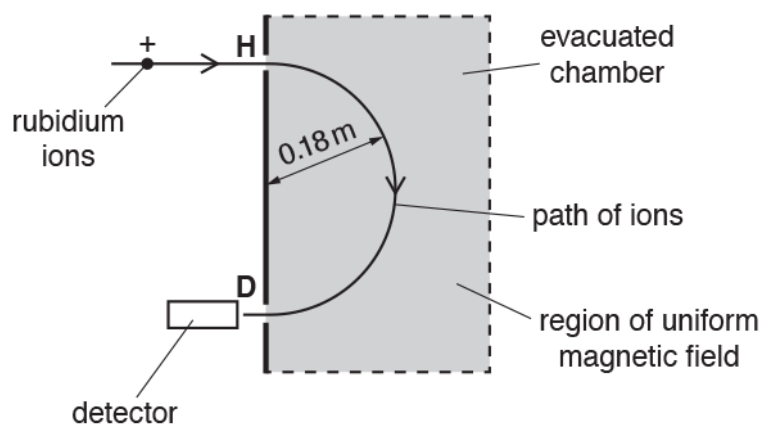


Fig. 20

A uniform magnetic field is applied to an evacuated chamber. The direction of the magnetic field is perpendicular to the plane of the paper.

A beam of positive rubidium ions enters the chamber through a hole at H. The ions travel in a semi-circular path in the magnetic field. The ions are detected at point D.

Each rubidium ion has charge $+1.6 \times 10^{-19} \text{ C}$ and speed $4.8 \times 10^4 \text{ m s}^{-1}$.

The radius of the semi-circular path of the ions is 0.18 m.

The mass of a rubidium ion is $1.4 \times 10^{-25} \text{ kg}$.

Calculate the magnitude of the magnetic flux density B of the magnetic field.

$B = \dots\dots\dots \text{ T [3]}$

8(a) Fig. 21.1 shows a coil of a simple generator rotating in a uniform magnetic field.

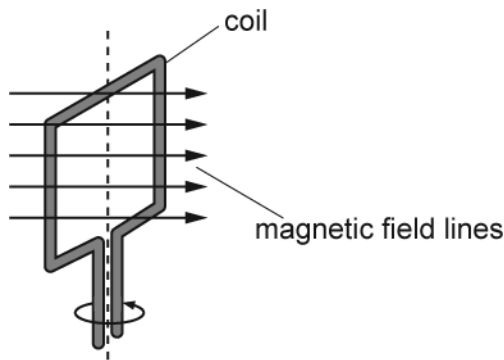


Fig. 21.1

The coil has 85 turns of insulated wire. The cross-sectional area of the coil is 14 cm².

Fig. 21.2 shows the variation of magnetic flux density B through the plane of the coil with time t as it rotates.

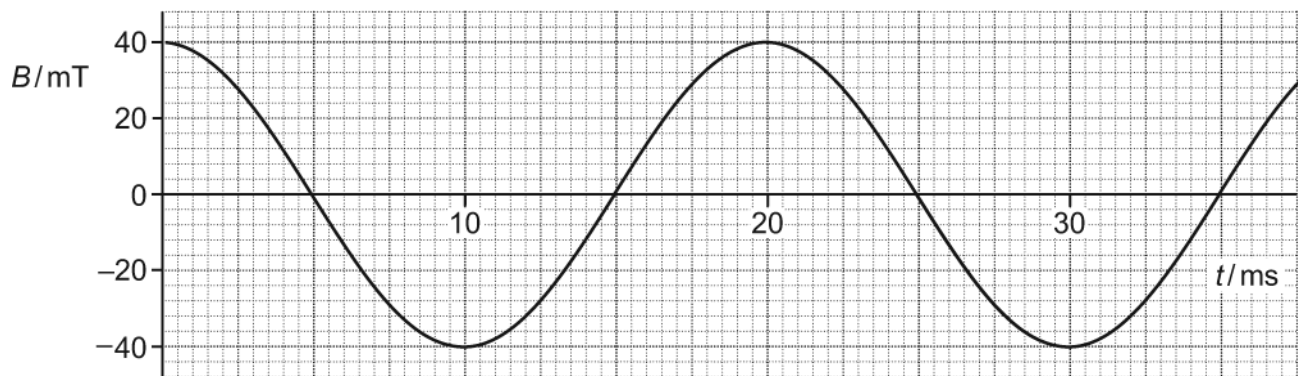


Fig. 21.2

- i. Explain why the electromotive force (e.m.f.) induced across the ends of the coil is a **maximum** at the times when $B = 0$.
-
-
- [1]
- ii. Draw a tangent to the curve in Fig. 21.2 when $B = 0$, and hence determine the **maximum** e.m.f. induced across the ends of the coil.

maximum e.m.f. = ----- V [3]

(b) Fig. 21.3 shows the variation of the e.m.f. induced across the ends of the coil with time t .

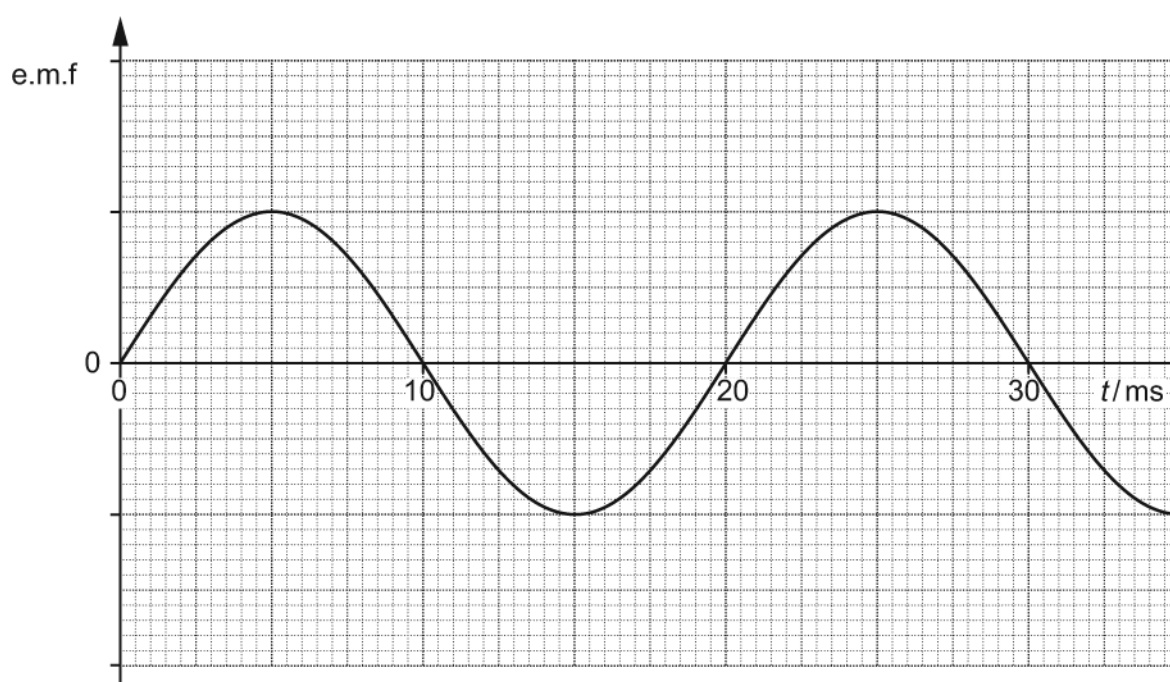


Fig. 21.3

The magnitude of the magnetic flux density of the uniform field is now halved and the coil is rotated at twice its previous frequency.

On Fig. 21.3 sketch the new variation of the e.m.f. induced with time t .

[2]

9(a)



A student conducts an experiment to confirm that the uniform magnetic flux density B between the poles of a magnet is 30 mT.

A current-carrying wire of length 5.0 cm is placed perpendicular to the magnetic field.

The current I in the wire is changed and the force F experienced by the wire is measured. Fig. 22.1 shows the graph plotted by the student.

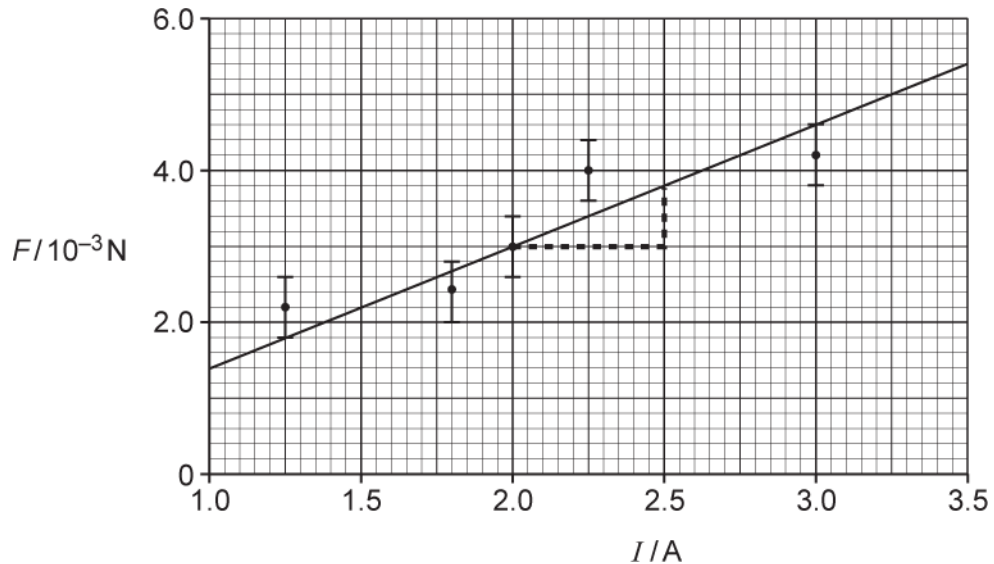


Fig. 22.1

The student's analysis is shown on the graph of Fig. 22.1 and in the space below.

$$F = BIL$$

$$\text{gradient} = BL = \frac{(3.8 - 3.0) \times 10^{-3}}{2.5 - 2.0} = 0.0016$$

$$B = \frac{0.0016}{0.05} = 0.032 \text{ T} = 32 \text{ mT}$$

This is just 2 mT out from the 30 mT value given by the manufacturer, so the experiment is very accurate.

Evaluate the information from Fig. 22.1 and the analysis of the data from the experiment. No further calculations are necessary.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school handwriting practice paper. The lines are evenly spaced and run across the width of the page. In the bottom right corner, there is a small black rectangular box containing the number [6].

The diagram shows a transformer with a rectangular soft-iron core. The primary coil is wound on the left vertical limb and is connected to an alternating supply. The secondary coil is wound on the right vertical limb and is connected to a filament lamp. Labels with leader lines identify the primary coil, secondary coil, soft-iron core, alternating supply, and filament lamp.



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- This image shows a blank sheet of white paper with eight horizontal dashed lines, typical of primary-ruled notebook paper. The lines are evenly spaced and extend across the width of the page. There is no handwriting or other markings on the paper.

ii. The primary coil has 400 turns and the secondary coil has 20 turns. The potential difference across the lamp is 12 V and it dissipates 24 W. The transformer is 100% efficient.

current = _____ A [2]

2. The alternating voltage supply is replaced by a battery and an open switch in series. The switch is closed. The lamp is lit for a short period of time and then remains off. Explain this observation.

----- [2]

- 10 A nucleus of hydrogen-3 (${}^3_1\text{H}$) is unstable and it emits a beta-minus particle (electron).

The emitted beta-minus particle enters a region of uniform magnetic field.

Fig. 22.1 shows the path of the particle **before** it enters the magnetic field.

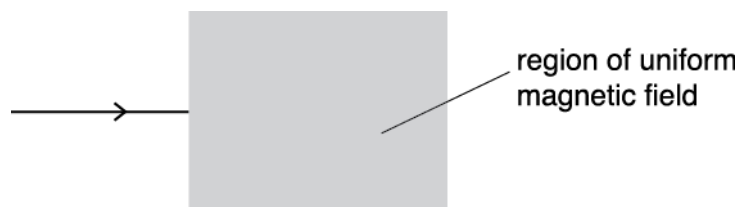


Fig. 22.1

The direction of the magnetic field is into the plane of the paper.

Describe and explain the path of the particle in the magnetic field.

----- [2]

- 11 A positively charged particle is travelling in a uniform field.
Fig. 21.1 shows the particle travelling at right angles to the direction of the field.

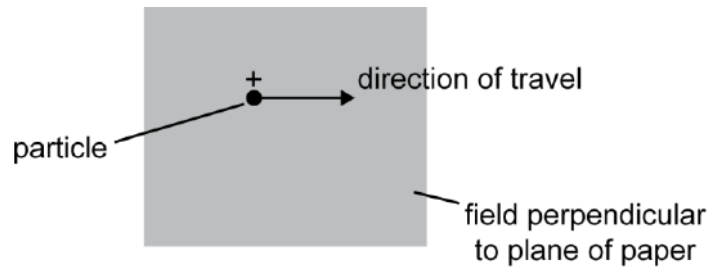


Fig. 21.1

Describe the motion of the particle in terms of the force it experiences when the field is

- i. a magnetic field

----- [2]

- ii. an electric field.

----- [2]

12(a) Fig. 22.1 shows the circular track of a positron moving in a uniform magnetic field.

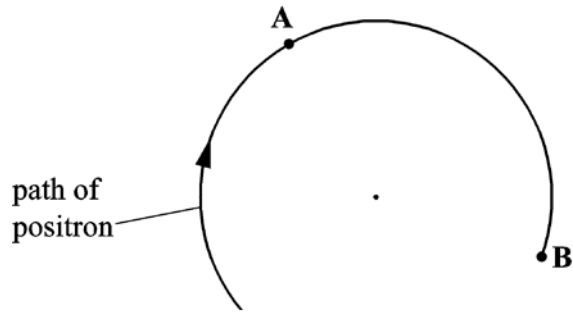


Fig. 22.1

The magnetic field is perpendicular to the plane of Fig. 22.1.

The speed of the positron is $5.0 \times 10^7 \text{ m s}^{-1}$ and the radius of the track is 0.018 m.

State the direction of the force acting on the positron when at point A and explain why this force does not change the speed of the positron.

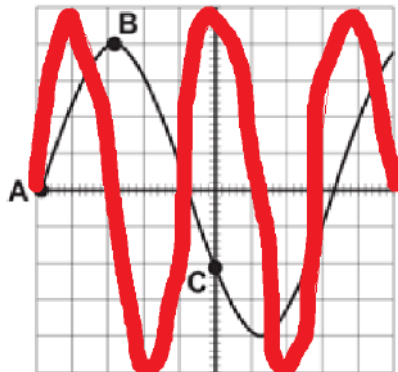
----- [2]

(b) Calculate the magnitude of the magnetic flux density of the magnetic field.

magnetic flux density = ----- T [3]

END OF QUESTION PAPER

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
1	a		B	A1	Examiner's Comments A very common incorrect response was A, presumably confusing flux linkage with induced e.m.f.
	b		Amplitude drawn is larger Time period drawn is smaller	B1 B1	At least one complete cycle needs to be drawn for any credit Consider all cycles drawn. All peaks must be greater than amplitude of given wave and consistent within $\frac{1}{2}$ square of each other For T, only mark first complete cycle – do not insist on symmetry but such that the end of the first cycle is less than or equal to 8 squares e.g.  Examiner's Comments The vast majority of candidates appreciated that an increased frequency results in a reduced time period and were able to show this on the graph. Not all were able to show a constant (reduced) time period, although this was not required on this occasion. Fewer candidates appreciated that the increased rate of change of flux linkage would produce a greater e.m.f. and drew the amplitude of the wave higher. The quality of wave drawing was not always good and care should be taken to aim to keep a consistent amplitude and time period in this sketch.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	c	i	$\Phi = BA \cos \theta = 25 \times 10^{-3} \times (8.0 \times 10^{-2} \times 4.0 \times 10^{-2}) \times \cos 45 (= 5.66 \times 10^{-5} \text{ Wb})$ Change in flux linkage = $N \Phi = 120 \times 5.66 \times 10^{-5}$ $= 6.8 \times 10^{-3} \text{ (Wb)}$	C1 A1	ALLOW $\sin \theta$ for $\cos \theta$ Only apply POT error in coil dimensions once Calculation of area of circle using one of the given coil dimensions is XP Correct to at least 2 significant figures (6.79×10^{-3}) IGNORE sign <u>Examiner's Comments</u> The terms flux and flux linkage can clearly be confused by candidates and they are often incorrectly interchanged. Many candidates simply calculated the flux. There were a significant number who calculated the area incorrectly, usually considering the appropriate area to be a circle. Several candidates attempted to calculate a change between two different angles, which resulted in an incorrect answer.
		ii	Induced emf ($E = - \Delta \Phi / \Delta t$) = (-) 0.0068 / 0.13 $\varepsilon = (-) 0.052 \text{ (V)}$	C1 A1	ALLOW ECF for flux linkage above but answer to (c)(i) $\times 120 \div 0.13$ scores zero marks for this question Correct to at least 2 significant figures IGNORE sign <u>Examiner's Comments</u> Those who obtained the correct response for part (c) (i) generally went on to obtain both marks in this question. Allowance of an "error carried forward" also helped many candidates, and a little over half were able to score both marks. Many candidates rounded their answer to 1 significant figure, thereby losing a mark. Candidates should always be encouraged to give their answers in keeping with the significant figures of the data provided in the question.
			Total	7	

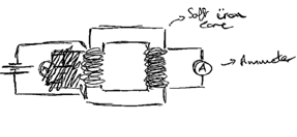
Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
2	a	Level 3 (5–6 marks) Clear description and clear analysis <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3–4 marks) Clear description or clear analysis or some description and some analysis <i>There is a line of reasoning presented with some structure. The information presented is in the most-part relevant and supported by some evidence.</i> Level 1 (1–2 marks) Limited description or limited analysis <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 mark No response or no response worthy of credit.	B1 × 6	Use level of response annotations in RM Assessor Indicative scientific points may include: Expect to see use of ac for clear description Description • labelled circuit diagram • smaller number of turns on primary • a.c. supply on primary • load resistor on output • a.c. ammeter to measure current • measurement of number of turns on secondary • keep number of turns on primary constant • keep input ac voltage constant • increase number of turns on secondary each time Analysis • algebra to get to $N_s \propto 1/I_s$ • graph drawn $1/I_s$ against N_s / I_s against $1/N_s$ • expect straight line through origin or • $\lg I_s$ against $\lg N_s$ • expect straight line to show gradient = -1 • therefore $N_s \propto 1/I_s$ or - if load resistor in circuit only • use of Ohm's law applied to load resistor to explain why presence of load resistor increases primary current • plotting graph drawn I_s against N_s • expect straight line through origin <u>Examiner's Comments</u> This Level of Response (LoR) question was designed to assess practical skills of planning, implementation, analysis and evaluation from Module 6 of the specification, specifically 6.3.3(f), along with HSW3 and HSW4. A holistic approach to marking is used, with marks given according to answers matching the

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			descriptors for the various levels. No one answer is perfect for this question, and examiners were expecting a varied approach, with common themes, which would lead to a correct conclusion. The nature of the question is such that it can be conveniently separated into a description of the experiment and an analysis of it. The key points in the description that examiners were looking for were: - a suitable circuit diagram with a transformer and coils along with a supply and metres - use of an ac supply - a description of the method used, specifically controlling variables, and measuring an output - a statement or description of how the output current is to be determined - a statement of how the secondary turns is to be varied. The key points in the analysis that examiners were looking for were: - a statement of the transformer equation in terms of current and turns - a suitable arrangement of this formula to give either the dependent or independent variable as the subject - a suitable graph which would give a linear relation - an explanation of how this graph would appear. It was clear that many candidates had carried out this experiment by investigating voltage however the use of current here caused some difficulties to less successful candidates. The very best responses were detailed, with a clear diagram and well-structured making every attempt to fully answer each section. In many cases the diagram did little to add to the description, however the very best were suitable on their own. Many candidates had a single supply for both the primary and secondary turns and a significant number used a dc supply. Good responses often included the additional detail that the transformer was a step up so that there would be a larger number of turns on the secondary, often giving suitable numbers. Many candidates stated that they would plot secondary current against secondary turns and then stated that it would show inverse proportionality. It is to be explained that any relation should be shown through a straight line

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
			graph wherever possible. Exemplar 2  Set up experiment as shown in diagram. Use a AC power supply on primary coil at constant voltage and an ammeter on other end. Wind the wire on each side keeping primary coil the same and noting down how many turns are on both coils. Switch on the power and take readings of the current from ammeter. After doing to increase the number of turns on secondary coil and repeat experiment. Repeat this at least 7 times, changing the number of turns by a constant interval each time. Once you have obtained all the data Plot a graph of Number of turns against the current in secondary coil. This should show a straight line through origin as $y = mx + c$ gradient $I_s = I_p \frac{n_p}{n_s} = I_p \times \frac{1}{n_s}$ should be constant and when there are no turns there is no induced E and $\therefore$ no current relationship $\therefore I_s \propto \frac{1}{n_s}$. Exemplar 2 shows a response which does not necessarily have all the indicative points contained but would be given Level 3 and 6 marks. Firstly, the diagram doesn't have any method for measuring input current or have a load resistor, but it is clear what it represents and it also looks like it is a step-up transformer. The diagram has a dc cell symbol, but this is overridden by a statement that it is an ac supply in the text. This does not have to be treated as a contradiction, as the symbol for ac is not likely to be known. The method explains how this is to be carried out quite well and is structured in an ordered way. The analysis is brief but complete. It shows the equation that has been used, arranged in the form of $y = mx + c$ and then stating what should be plotted and what it will show. This is an example of a perfectly good response that will achieve maximum

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					marks without having to get every indicative point.
	b	i	The (magnitude of the induced) e.m.f. is (directly) proportional/equal to the rate of change of (magnetic) flux linkage	B1	Allow electromotive force for e.m.f. Do not allow electromagnetic force for e.m.f. Allow change per unit time for rate of change Allow cutting flux for change of flux linkage Do not allow flux density for flux linkage <u>Examiner's Comments</u> It was encouraging to see that over half of the candidates were able to correctly recall Faraday's law, which is specifically stated in the specification. The vast majority of incorrect responses were a statement, generally in words, of the transformer equation presumably as it was used in the following question.
		ii	<u>Use of</u> $N_s / N_p = V_s / V_p$ $N_s / 920 = 5.0 / 230$ so $N_s = 20$	C1 A1	Expect to see attempt at substitution following the equation Correct substitution and correct algebra into any arrangement <u>Examiner's Comments</u> The vast majority of candidates were able to correctly show the number of turns on the secondary coil was 20. As always with a 'show that' question it is important that each step is clearly shown and that there is no doubt that the candidate has evaluated the answer. Candidates who did not score on this generally were not explicit enough in their working.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		iii	Use of $N = 20$ $3.4 = 20 \times \Delta\phi / 1.2 \times 10^{-3}$ $\Delta\phi = 2.0 \times 10^{-4}$ Unit: Wb / weber(s)	C1 C1 A1 B1	Expect to see factor of 20 correctly used Correctly calculated to 2sf (2.04) Special cases: for two marks from the calculation Use of $N = 1$ gives 4.08×10^{-3} correct to 2sf Use of $N = 920$ gives 4.43×10^{-6} correct to 2sf Allow $T \text{ m}^2$ Ignore $V \text{ s}$ Allow correct base unit ($\text{kg m}^2 \text{ s}^{-2} \text{ A}^{-1}$) Examiner's Comments This question was generally well answered with over a third gaining all 4 marks. Most candidates chose the correct equation, although its rearrangement caused some problems, along with the number of turns to use. The correct numerical answer was 2.0×10^{-4} which several candidates incorrectly (in terms of significant figures) gave as 2×10^{-4} thus losing a mark. Candidates should be reminded that, in general, answers should be given to the lowest number of significant figures in the question, in this case 2sf. Many candidates who struggled with the calculation were able to get the standalone mark for the unit although there was a reasonable number of W rather than Wb.  Misconception The formula stating $\Delta(N\Phi)$ seemed to cause some misunderstanding and a fair number of candidates seemed to calculate the difference in the number of turns ($920 - 20$) therefore using 900 as the number of turns.
			Total	13	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
3	a	i	Oscilloscope / CRO	B1	Allow <u>a.c.</u> voltmeter Ignore datalogger / multimeter Examiner's Comments Most candidates appreciated that a voltage was to be measured, however it was evident that only a small proportion realised that it would be alternating and so a simple "voltmeter" would not be sufficient. Candidates putting the correct response of "oscilloscope" may well be those who have carried out practical work using search coils.
		ii	1 f 2 θ or $\sin \theta$	B1 B1	Not any other symbol. Only mark quantity letters – ignore any words, but allow frequency. Allow θ or $\sin \theta$ with any or all of K, I_0, A, N. Only mark quantity letters – ignore any words. Examiner's Comments Part 1 was answered better than part 2 in general. For part 1, the majority of candidates appreciated that the "rate" will have included the frequency although many included other irrelevant (and therefore incorrect) terms too. In part 2, the concept of what causes the magnetic flux linkage to "change" did not appear to be well understood; A and/or N were often an incorrect response, presumably as the candidate was aware that these terms are included in a calculation for flux linkage.

Mark Scheme

Question	Answer/Indicative content	Marks	Guidance
	iii $f = \frac{0.62}{5000 \times 4.0 \times 10^{-3} \times 8.0 \times 7.8 \times 10^{-5}} \quad / f = 49.67 \dots \text{ (Hz)}$ $\frac{0.2}{8.0} \quad / \quad \frac{0.1}{7.8} \quad / \quad \frac{0.03}{0.62}$ $\text{abs uncertainty} = \left(\frac{0.2}{8.0} + \frac{0.1}{7.8} + \frac{0.03}{0.62} \right) \times 49.67 \dots / 4.28 \dots \text{ (Hz)}$ $f = 50 \pm 4 \text{ (Hz)}$	C1 C1 C1 A1	Any individual raw uncertainty Max value = 54.11 (Hz) and min value = 45.54 (Hz) for f Allow 8.6% as evidence of this calculation For min / max method: difference / 2 = 4.29 (Hz) Allow ecf on abs uncertainty from incorrect f Any ecf on f must be given to 2sf and uncertainty sf consistent. Not the paper SF penalty <u>Examiner's Comments</u> A good fraction of candidates were able to score full marks on this question. It is clear that many had been well prepared in treatment of errors, and 8.6% was seen often in the working. A common mistake among more successful responses was giving the error as 4.3, rather than 4. Less successful often simply added the raw uncertainties, giving 0.33, which was often then placed on the answer line. Some candidates missed out the factor of 10^{-5} in their calculation of f. Other approaches to obtain errors, such as calculating maximum and minimum values for f were seen and these can also lead to full marks.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	When switch is opened, there is (rate of) change in (magnetic) flux (linkage) which induces an emf / current as the (magnetic) flux links to B which causes the lamp to light (Lamp off) at start / end there is constant flux / no change in (magnetic) flux (linkage for coil B)	B1 B1 B1	Not just a statement of Faraday's law Allow flux is cut by B <u>Examiner's Comments</u> Many candidates found difficulties in appreciating what was required. A major confusion arose from a misunderstanding of the actual process, in that it was the opening of the switch which caused the lamp to light. Many knew Faraday's law, and were able to quote it, but not able to put it correctly into the context of this question. A common error included a misunderstanding of the role of the rod. With a large number of less successful responses stating that it conducted current to the lamp. A large number of candidates did not discuss the process in the terms of magnetic flux (as was required in the question) but talked in vague terms about "magnetism". Despite this being a challenging question, many candidates were able to score marks, and those at the top end were able to give clear and well-structured responses.
		Total	10	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
4			Apply a magnetic field at right angles to electric field electric force = magnetic force No resultant vertical force, so only beta-particles with a specific speed will travel horizontally	B1 B1 B1	Note this mark is for the idea that E and B are <u>perpendicular</u> even if direction of B is incorrect Allow 'apply horizontal magnetic field' Allow $Eq = Bqv$ Allow $v = E/B$ in this arrangement <u>Examiner's Comments</u> This question was quite poorly answered, with many candidates not even mentioning the magnetic field. Few appreciated that the magnetic field needs to be placed perpendicularly to the electric field, although most could state the $EQ = BQv$. However, in a description, there was some confusion about the 'fields' being equal rather than the 'forces'. No candidate gave a suitable description for the last mark but could access it through use of $v = E/B$.
			Total	3	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
5	a	Direction of field shown as clockwise	B1	Expect at least one field line with an arrow
		<u>Three</u> field lines shown as concentric circles and distance between adjacent field lines increasing as distance from wire increases	B1	Allow more than three lines, but distance between adjacent field lines increasing distance from wire must increase for all <u>Examiner's Comments</u> This question requires the candidates to identify the direction of the field and also to appreciate that the magnitude of the field reduces as the distance from the wire increases. Only around half were able to apply the right hand rule correctly to determine the direction, and only around 10% scored both marks. The increasing separation of the field lines with distance was poorly done for the most part. Many candidates kept the same separation, however those that may have attempted to increase this did not do with any clarity, so that parts of the circle would decrease. In general, the quality of the circles meant that it was difficult to be sure what the candidate's intention was. Some candidates were confused by the leader line, thinking it was the wire and attempted to draw a pattern around this. The question is clear that the diagram represents a top-view.

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
	b	(force =) $2.2 \times 10^{-3} \times 9.81$ $2.2 \times 10^{-3} \times 9.81 = B \times 5.0 \times 0.060$ (= 0.072 T) (absolute uncertainty =) $\frac{0.2}{6.0} + \frac{0.1}{5.0}$ ($\times 0.072$ = 0.0038 T) $B = 0.072 \pm 0.004$	C1 C1 C1 A1	Allow calculation of percentage uncertainty = 5.3% Allow calculation of max B (=0.0759 T) and min B (=0.0683 T) Note <i>B</i> must be given to 2 SF and the uncertainty given to 1 SF. Special case: allow follow through from incorrect B calculation. <u>Examiner's Comments</u> This question is based around a common experiment used to determine the magnetic flux density of a pair of magnets and the experimental design should have been familiar to many candidates, along with the use of $F = BIL\sin\theta$ from the data booklet. The first mark is for identifying the magnitude of the force as being the change in the apparent weight on the balance. Several candidates simply used the reading with the wire, or did not change the mass unit to kg. However, those who managed to get the correct reading for the force generally went on to calculate the magnetic flux density correctly. The uncertainties for two readings were given, and most candidates correctly calculated a percentage uncertainty of 5.3%. The final answer required the correct number of significant figures. Some candidates either did not see this, or ignored it, leaving their final answer in different significant figures. It was noted that several candidates underlined this instruction and in general they tended to follow it. It is good practice to do this.
		Total	6	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
6	a	i	(induced) e.m.f. is (directly) proportional / equal to the rate of change of (magnetic) flux linkage	B1	Not current Allow 'rate of cutting' for 'rate of change'
		ii	Connect the primary (coil) to an alternating voltage / current Oscilloscope connected across secondary coil / to measure E A graph of E against N will be a straight line through the origin.	B1 B1 B1	Allow AC (can be on the figure) Not changing / variable for alternating Allow voltmeter (can be on the figure) Allow p.d. / voltage for e.m.f. / E throughout Ignore any component (e.g. lamp or resistor) connected across the secondary coil Allow $(E \div N) = \text{constant}$
	b		Magnetic field (around current-carrying wire) (Fleming's) left-hand rule mentioned (Magnetic) field into page, (current is up the page) and force is to the left / towards X	B1 B1 B1	Not magnetic force Allow 'field into page and wires attract' Note the field direction and force direction can be shown on the figure
			Total	7	
7			$F = BQv$ and $F = mv^2/r$ or $B = mv/Qr$ (Any subject) $(B =) \frac{1.4 \times 10^{-25} \times 4.8 \times 10^4}{1.6 \times 10^{-19} \times 0.18}$ $B = 0.23 \text{ (T)}$	C1 C1 A1	Allow e <u>Examiner's Comments</u> This question on the circular motion of charged particles in a uniform magnetic field was answered with confidence and flair. Most candidates got the correct answer of 0.23 T for the magnetic flux density. A small number of candidates, mainly at the low-end, were using incorrect equation for the magnetic force experienced by the ions. Some of these equations were hybrids of the electric force experienced by charged particles.
			Total	3	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
8	a	i	The gradient is maximum / maximum rate of change of B / maximum rate of change of flux (linkage)	B1	Allow slope instead of gradient <u>Examiner's Comments</u> Although worth just 1 mark, this question did provide good opportunity for top-end candidates to pick up one mark. Many candidates quoted Faraday's law of electromagnetic induction, without mentioning that the rate of change of flux (linkage) was maximum at $B = 0$. Low-scoring candidates wrote about the orientation of the coil relative to the magnetic field or the '<i>cutting</i>' of field lines. None of the explanations led to any marks being credited.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	Tangent drawn to curve at $B = 0$ gradient = 12.5 (maximum e.m.f. = $12.5 \times 14 \times 10^{-4} \times 85$) maximum e.m.f. = 1.5 (V)	C1 C1 A1	Allow 11.70 to 13.30; no need to check calculation Allow fraction if calculated value is within the range Allow ECF from the gradient value if value is outside the range Alternative: $E = BAN_{\omega}$ C1 $E = 40 \times 10^{-3} \times 14 \times 10^{-4} \times 85 \times 2\pi \times 50$ C1 maximum e.m.f. = 1.5 (V) A1 Examiner's Comments Most candidates followed the question and drew decent tangents on Fig. 21.2. Most of the tangents were acceptable, but a few either crossed the curve or had very thick pencil lines. A significant number of candidates quoted the maximum e.m.f. to be equal to the magnitude of the gradient of the tangent. Top-end candidates faced no obstacles here; the gradient was multiplied by $[85 \times 14 \times 10^{-4}]$ to give an answer around 1.5 V. Once again, a good number of candidates were picking the odd mark through error carried. Converting the cross-sectional area of 14 cm^2 into $14 \times 10^{-4} \text{ m}^2$ was a challenge for some of the candidates in the middle and lower quartiles.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
	b		Sinusoidal curve with the same peak e.m.f. Sinusoidal curve with half period	B1 B1	Note curve must show at least half a period Allow ± 1 small square for e.m.f. Ignore phase Note graph must show at least half a period Allow ± 1 small square for t <u>Examiner's Comments</u> Most candidates scored a mark for showing that the period of the new e.m.f. trace was halved. Only a small proportional had the peak e.m.f. unchanged; the most frequent incorrect trace showed the peak e.m.f. also being halved. The sinusoidal curves were generally well-sketched.
			Total	6	

Mark Scheme

Question		Answer/Indicative content	Marks	Guidance
9	a	Level 3 (5-6 marks) Clear evaluation of Fig. 22.1 and clear analysis <i>There is a well-developed line of reasoning which is clear and logically structured. The information presented is relevant and substantiated.</i> Level 2 (3-4 marks) Some evaluation of Fig. 22.1 and some analysis <i>There is a line of reasoning presented with some structure. The information presented is in the most part relevant and supported by some evidence.</i> Level 1 (1-2 marks) Limited evaluation of Fig. 22.1 or limited analysis <i>There is an attempt at a logical structure with a line of reasoning. The information is in the most part relevant.</i> 0 marks <i>No response or no response worthy of credit.</i>	B1×6	Use level of response annotations in RM Assessor, e.g. L2 for 4 marks, L2⁺ for 3 marks, etc. Ignore incorrect references to the terms precision and accuracy Indicative scientific points may include: Evaluation of Fig. 22.1 • Comment on the line • The straight line misses one error bar / anomalous point ringed or indicated • Too few data points plotted • The triangle used to calculate the gradient is (too) small • Some plots should have been repeated / checked • No error bars for current • 'Not regular intervals' (for current) • No origin shown (AW) Evaluation of analysis • The value of B is close to the accepted value • The difference of only 7% • No absolute or percentage uncertainty in B shown (AW) • Worst-fit line or maximum / minimum gradient line could have been used to determine the (absolute or percentage) uncertainty in B • F against I graph should be a straight line or • $BL = \text{gradient}$ (any subject) Examiner's Comment This was the second level of response (LoR) question in the paper. It required evaluation of a graph drawn by a student and the analysis shown in the box on page 24. Most candidates realised that the graph had few data points, the triangle used for the gradient was too small and the line drawn totally missed one of the error bars. The analysis shown by the candidate did not include an absolute uncertainty in B, which made the statement written by the student lack credibility. Many candidates wrote about drawing doing a line of worst-fit and determining the percentage uncertainty. This was only possible if there were more

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
					data points and the error bars for the F values reduced by perhaps repeating the measurements. Once again, there was a good spread of marks amongst the three levels.
	b	i	There is a changing / fluctuating (magnetic) field / flux (linkage) (magnetic) field / flux (linkage) in <u>core</u> and <u>secondary</u> (coil) Statement of Faraday's law: e.m.f. (induced) $\propto$ <u>rate</u> of change of (magnetic) flux <u>linkage</u>	M1 A1 B1	Note: This changing flux can be anywhere Allow 'the direction of the field oscillates' Allow 'the core helps to link the flux to the secondary coil' Allow 'equal to / =' Ignore 'cutting of flux' Not just $E = (-)\Delta(N\phi)/\Delta t$ Examiner's Comment The topic electromagnetic induction always challenges candidates. Successful responses often showed correct use of technical terms such as <i>magnetic flux</i> or <i>flux linkage</i>. Most candidates scored a mark for correctly stating Faraday's law of electromagnetic induction. Many realised that an alternating current produced an alternating magnetic flux within the iron core and this change in flux produced an e.m.f. at the secondary coil. One of the popular misconceptions was that there was an alternating current (or induced e.m.f.) within the iron-core. A small number of candidates referred to electromagnetic field in their descriptions rather than magnetic field.

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
		ii	1 ($I_S =$) 24/12 or 2.0 (A) $(I_P =) \frac{20}{400} \times 2.0$ (current in primary =) 0.10 (A) or $(V_P =) 12 \times 20$ or 240 (V) $(I_P =) \frac{24}{240}$ (current in primary =) 0.10 (A) 2 Idea of changing / increasing (magnetic) field / flux / current (in primary) at the start Eventually current and flux (linkage) are constant, therefore no e.m.f.	C1 A1 C1 A1 B1 B1	Allow 1 sf answer Allow 1 sf answer Note: Any labels used must be clearly defined Examiner's Comment This question on current in the primary coil was successfully answered by most candidates. The most favourable method was to calculate the current in the secondary and then the current in the primary coil. The turn-ratio equation and $P = VI$ were effortlessly used to arrive at the correct answer of 0.10 A. Full marks were rarely scored but many top-end candidates did manage to score a mark for suggesting that the lamp was lit for a short period of time at the start because '<i>there was a changing magnetic flux as the current increased from zero to a steady value</i>'. Too many answers focussed on the requirement of an alternating supply for an induced e.m.f. in the secondary coil and how a battery is not an alternating supply.
			Total	13	

Mark Scheme

Question			Answer/Indicative content	Marks	Guidance
10			Flemings left hand rule / the force on the electron is in the plane of the paper, right angles to the velocity and 'downwards'.	B1	Note: If drawn on Fig. 22.1, then judge 'circular' path by eye.
			Circular path within field in a clockwise direction.	B1	
			Total	2	
11		i	The force is right angles to the motion / velocity.	B1	
		i	The particle describes a circle in the plane of the paper.	B1	
		ii	Particle experiences a force perpendicular to motion / velocity.	B1	
		ii	It moves to the right and either comes out or goes into the plane of the paper (in a parabolic path).	B1	
			Total	4	
12	a		The force is towards the centre of the circle.	B1	
			The force is perpendicular to the motion or no component of force in direction of motion; hence no work is done on the particle.	B1	
	b		centripetal force provided by BQv ; hence $\frac{mv^2}{r} = BQv$	C1	
			$B = \frac{mv}{Qr} = \frac{9.11 \times 10^{-31} \times 5.0 \times 10^7}{1.6 \times 10^{-19} \times 0.018}$	C1	
			$B = 1.6 \times 10^{-2}(\text{T})$	A1	
			Total	5	